SALES DATA ANALYSIS

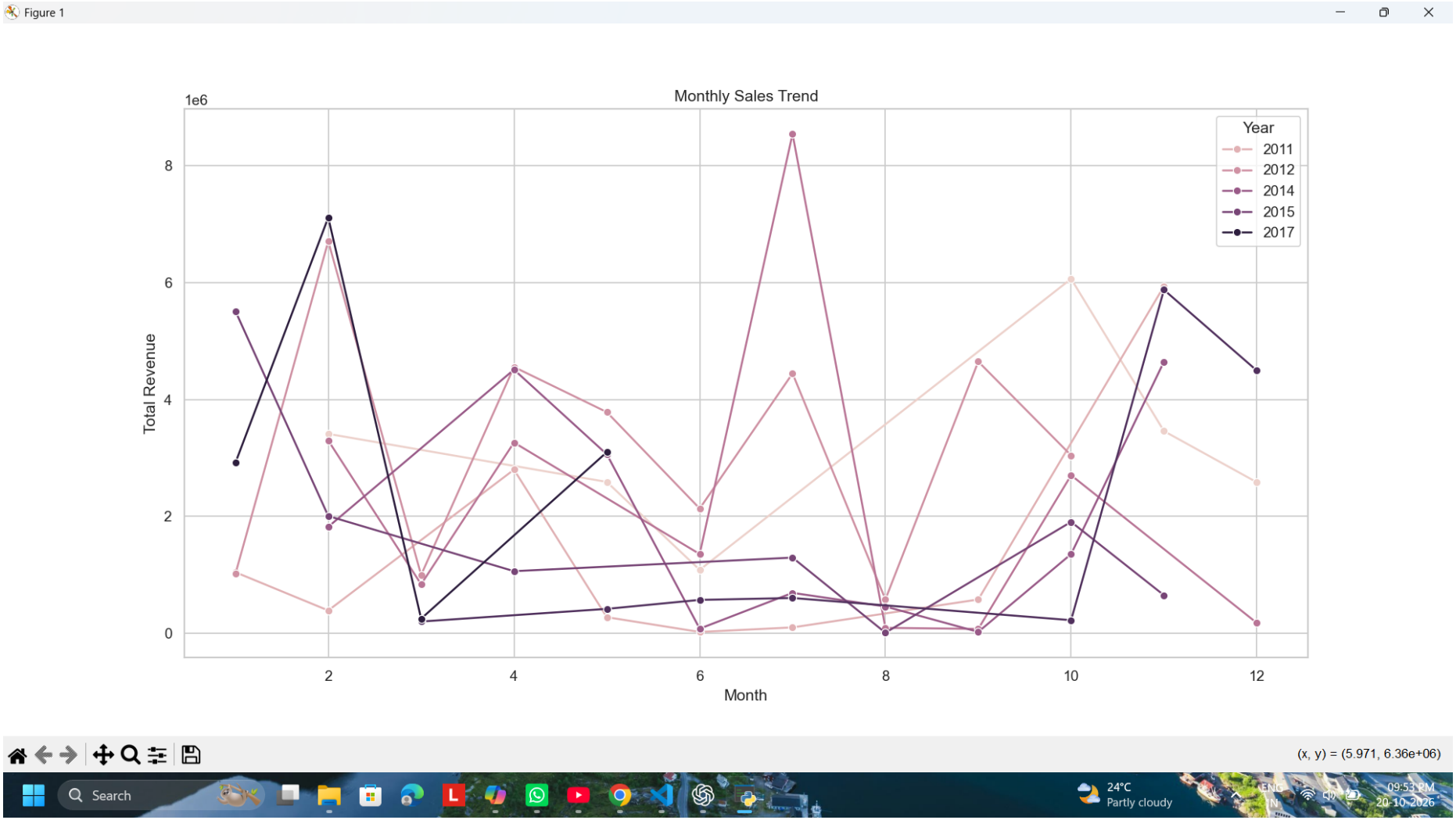
Project Results & Data Visualizations

This report presents the output charts generated by the Python sales analysis project. It includes the monthly sales trend, category-wise revenue, units sold by product, and monthly revenue heatmap.

Analysis area	What the chart shows
Monthly Sales Trend	Revenue by month, with separate lines for each year represented in the dataset.
Total Sales by Category	Total revenue aggregated for each item type.
Units Sold by Product	Total quantity sold in each item type, sorted from highest to lowest.
Monthly Revenue Heatmap	Revenue by year and month; darker/lighter cells encode different revenue levels.

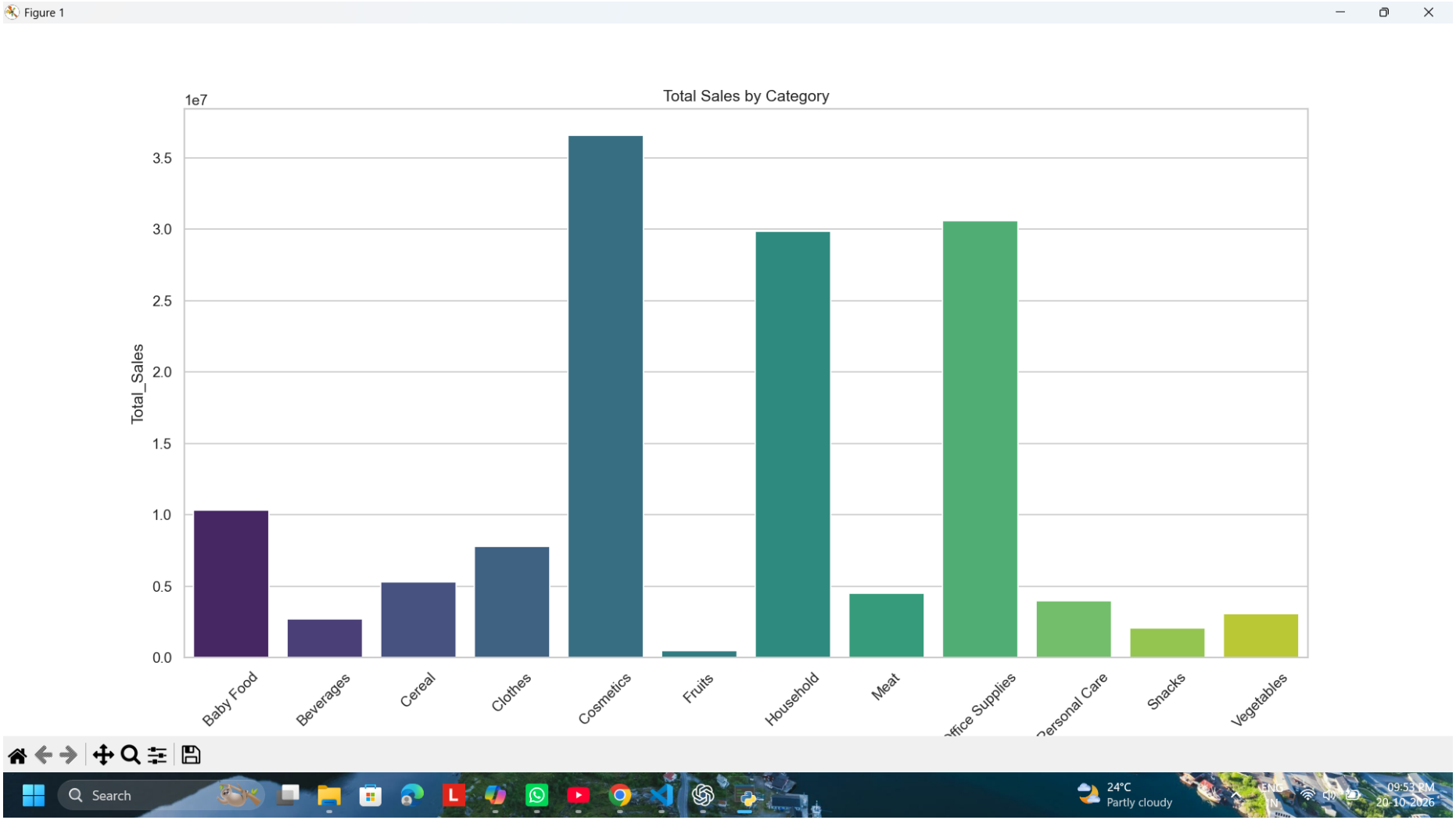
1. Monthly Sales Trend

Compares monthly revenue across the years shown in the dataset. Peaks and dips help identify unusually strong or weak sales months.



2. Total Sales by Category

Compares total revenue for each item type. The tallest bar represents the category with the highest total revenue in this dataset.



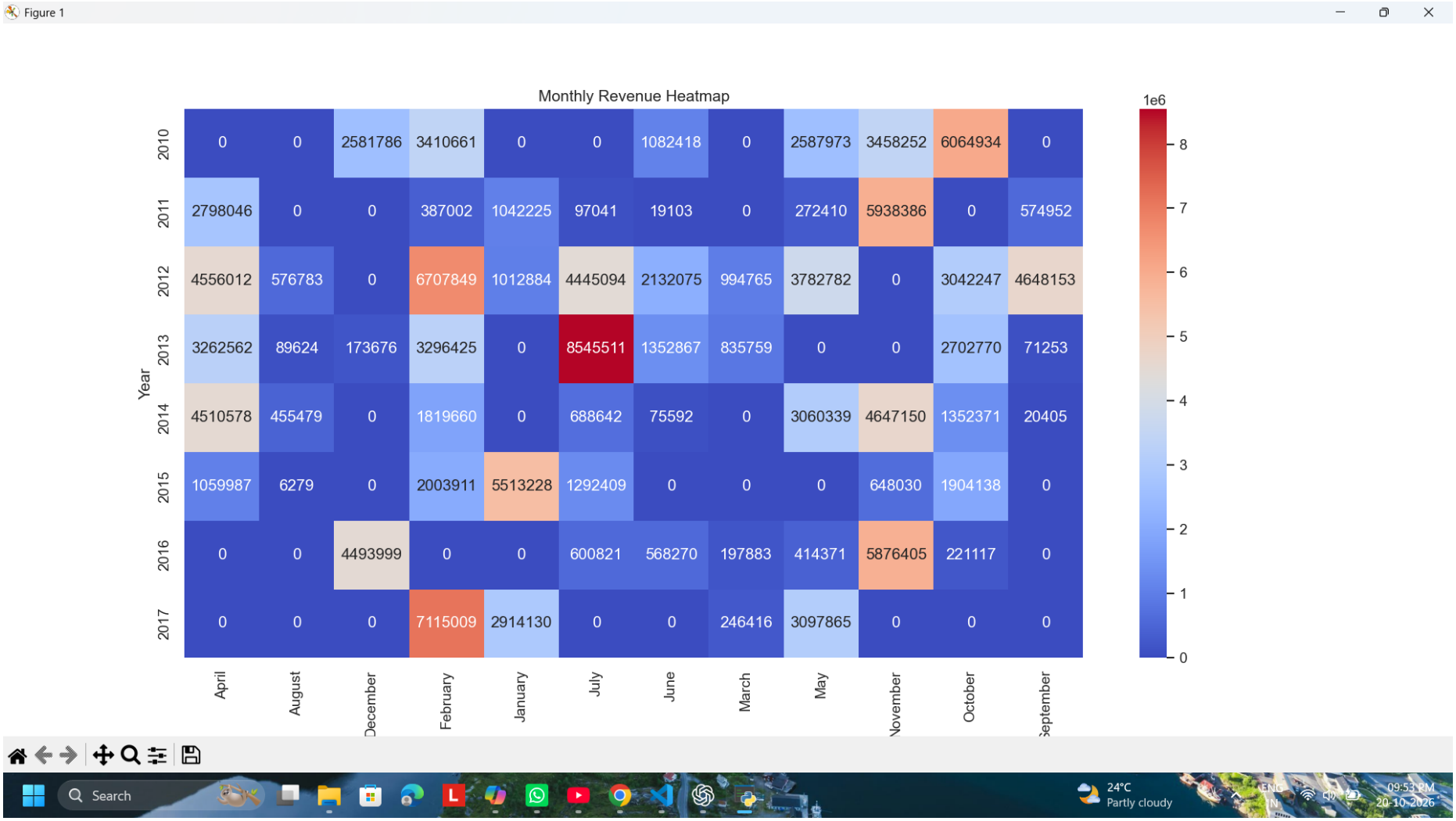
3. Units Sold by Product

Shows total units sold for each item type, ordered from most units to fewest units.



4. Monthly Revenue Heatmap

Shows the distribution of revenue across years and calendar months. Cells displayed as zero indicate no recorded revenue for that year-month combination in the plotted data.



Key Findings from the Printed Analysis

Metric	Value from the run output
Transactions after duplicate removal	98
Unique item types	12
Total revenue	137,348,768.31
Total units sold	512,871
Highest-revenue category	Cosmetics — 36,601,509.60
Lowest-revenue category	Fruits — 466,481.34
Highest monthly revenue shown in output	July 2013 — 8,545,511.20
Lowest monthly revenue shown in output	August 2015 — 6,279.09

Note: The monthly trend chart plots month number (1–12) on the x-axis and uses year as the hue. The heatmap currently uses month names as columns, which may appear in alphabetical order rather than calendar order. For a more polished analysis, month columns can be explicitly ordered January through December.